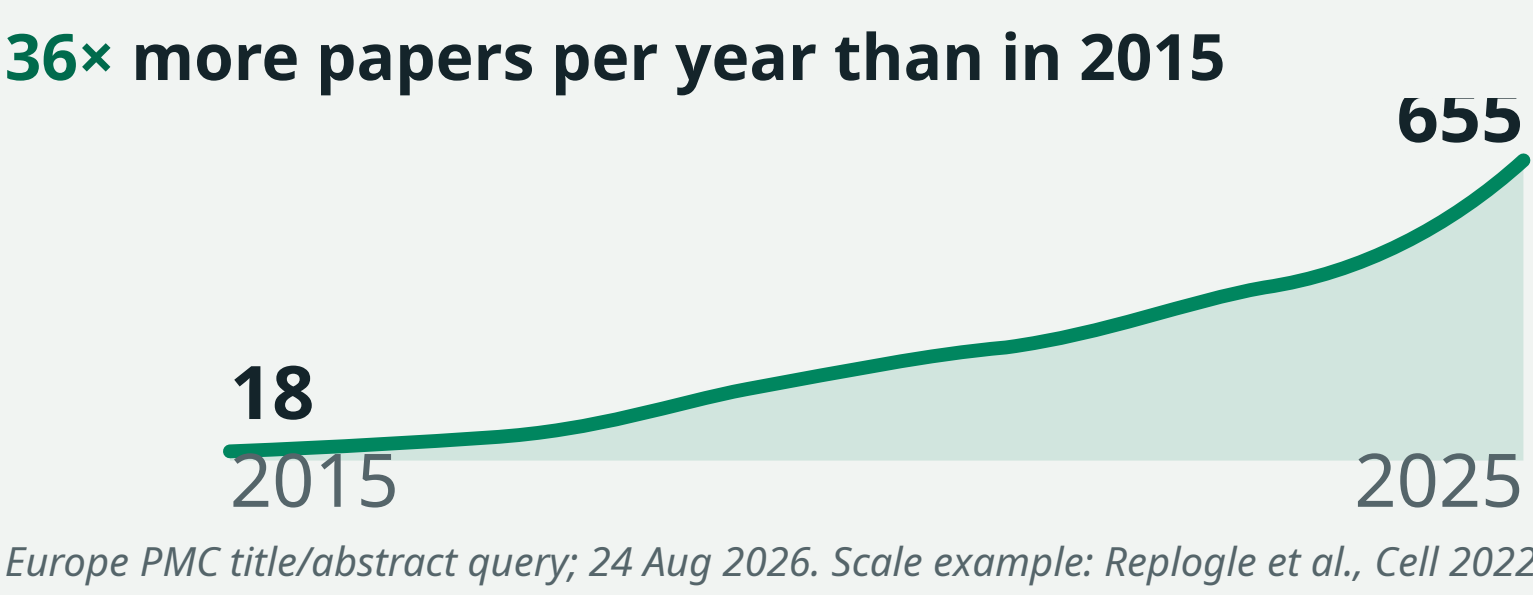
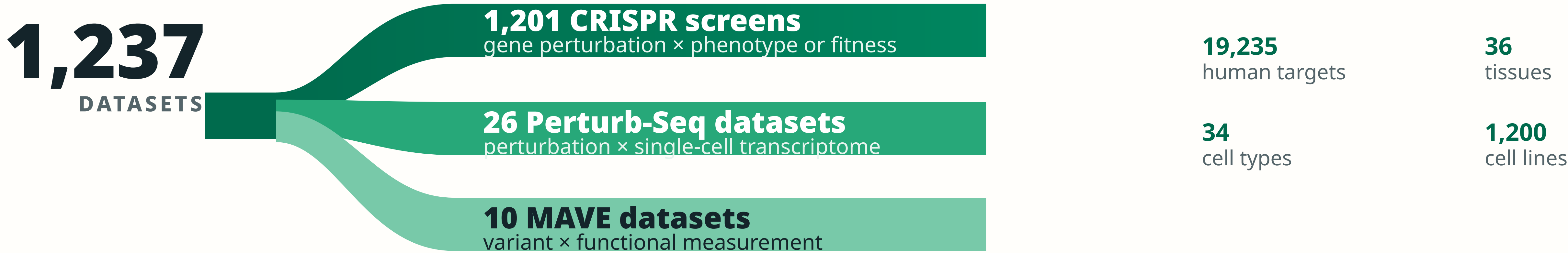


Perturbation evidence is multiplying faster than it can be organised

A genome-scale Perturb-Seq study can profile **more than 2.5 million cells**, and pooled screens can test thousands of targets in one experiment. That scale promises the statistical power to distinguish robust biology from study-specific noise—but the evidence is split across repositories, identifiers, assay conventions, reference genomes and incompatible analysis choices. Without a shared account of what was perturbed, in which model, with which readout and under which pipeline, combining studies remains slow, fragile and difficult to reproduce. The Catalogue makes that cross-study power practical: one curated schema, one stable target identity and one access path for comparing results.



A single catalogue turns scattered evidence into a searchable resource



Metadata and reanalysis make results comparable

84 searchable fields in the dataset contract

Study & provenance
Title, authors, year, source URI, licence, linked datasets

Model system
Organism, tissue, cell type or line, disease, sex, stage

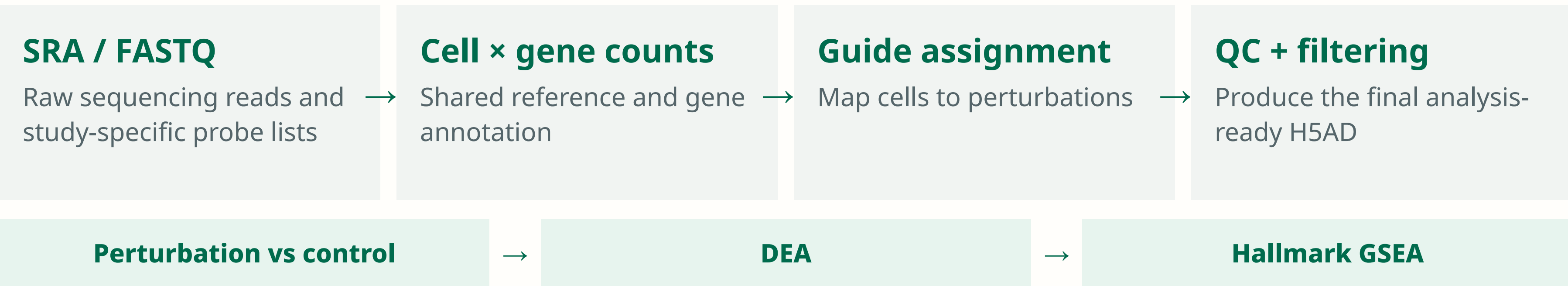
Perturbation & library
Targets, library design, delivery, guides, variants

Readout & processing
Assay, technology, sequencing kit, reference, software

Stable Ensembl gene IDs and ontology-linked entities make the same biological question addressable across modalities.

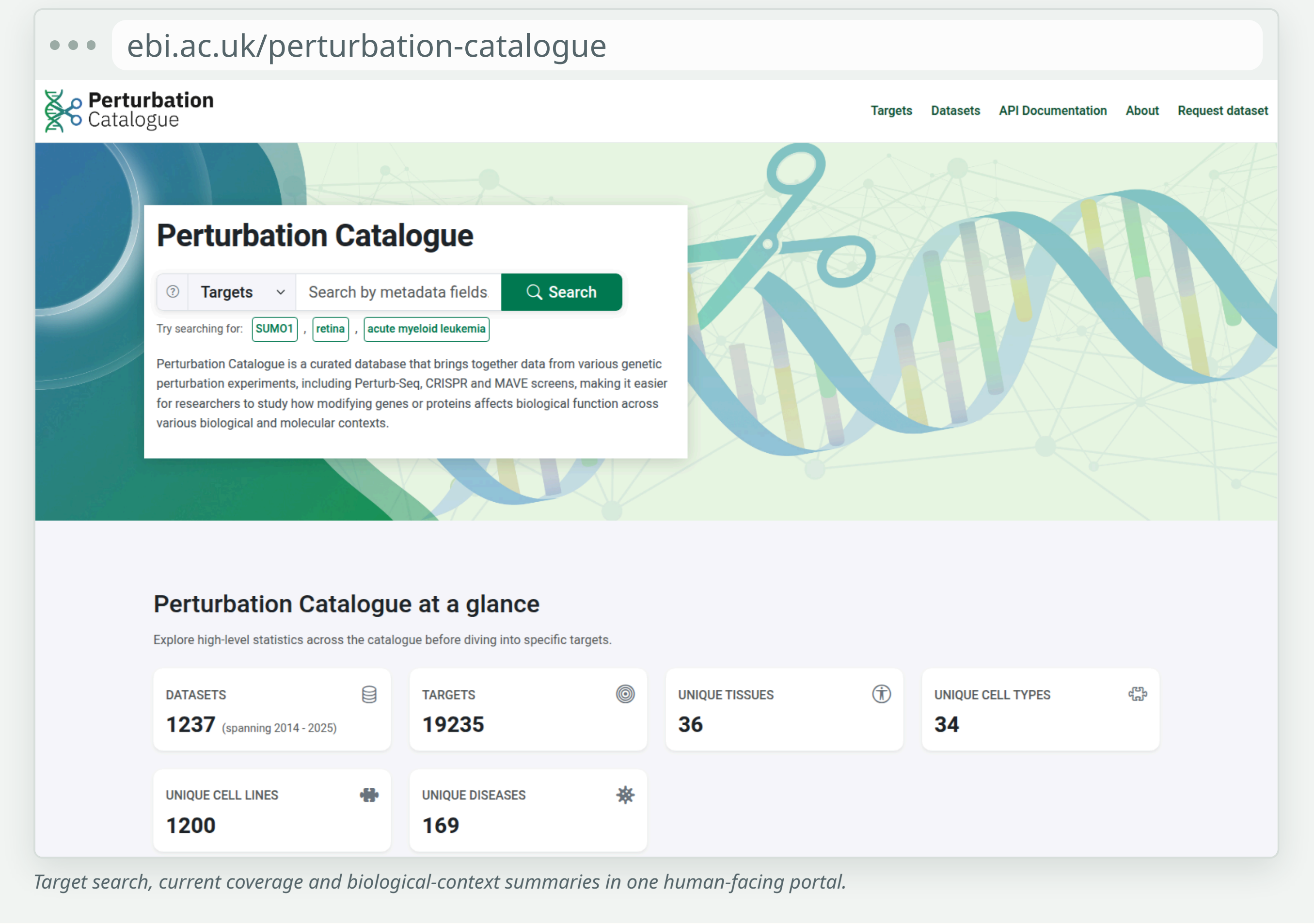
Perturb-Seq reprocessing starts again from raw reads

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.



Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Explore the Catalogue in a browser



Target search, current coverage and biological-context summaries in one human-facing portal.

Every result is programmatically accessible

Search, metadata, modality results and complete downloads use the same production backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/nadig_2025_jurkat
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
GET /v1/perturb-seq/nadig_2025_jurkat/download?format=csv.gz
```

```
{
  "total_rows_count": 30078802,
  "results": [{
    "perturbation": {
      "gene_symbol": "TFAM",
      "perturbed_target_ensg": "ENSG00000108064"
    },
    "effect": {
      "gene_symbol": "MT-ND5",
      "log2fc": -1.926,
      "padj": 0.0,
      "direction": "decreased"
    }
  }]
}
```

Next, integration becomes continuous and the analytical foundation becomes stronger

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.

Bring us the next hard perturbation-data problem
Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations around your programme's modalities, analyses and functionality.
perturbation-catalogue-help@ebi.ac.uk

Explore the Catalogue

Industry and research collaboration



